

Integration - Set 1 - Solutions

Specification MM7.1-MM7.6.

Answer key

Question	Answer	Topic
1	F	Integration
2	C	Integration
3	C	Integration
4	B	Integration
5	D	Integration
6	E	Integration
7	F	Integration
8	A	Integration
9	C	Integration
10	E	Integration
11	D	Integration
12	C	Integration

Worked solutions

Question 1 INT-02 • Integration • spec MM7.1

Find the area enclosed between the curve $y = x^2 - 4x$ and the x -axis.

Answer: F

Roots 0 and 4; the curve is below the axis between them. $\int_0^4 (x^2 - 4x) dx = \frac{64}{3} - 32 = -\frac{32}{3}$, so the area is $\frac{32}{3}$.

Question 2 INT-03 • Integration • spec MM7.3

Given that $\int_1^5 f(x) dx = 7$ and $\int_3^5 f(x) dx = 2$, find $\int_1^3 (2f(x) - 1) dx$.

Answer: C

$\int_1^3 f = 7 - 2 = 5$. So $2 \times 5 - (3 - 1) = 8$.

Question 3 INT-04 • Integration • spec MM7.6

A curve has $\frac{dy}{dx} = 6x^2 - 4x + 1$ and passes through $(1, 3)$. Find its equation.

Answer: C

$y = 2x^3 - 2x^2 + x + c$; at $(1, 3)$: $2 - 2 + 1 + c = 3 \Rightarrow c = 2$.

Question 4 INT-06 • Integration • spec MM7.5

The trapezium rule with 2 strips of equal width is used to estimate $\int_0^2 x^2 dx$. Which of the following is true?

Answer: B

$h = 1$: $\frac{h}{2}[y_0 + 2y_1 + y_2] = \frac{1}{2}[0 + 2(1) + 4] = 3$. The true value is $\frac{8}{3}$; the curve bends upwards (convex), so the chords lie above it and the rule overestimates.

Question 5 INT-07 • Integration • spec MM7.3

Given that $\int_0^a (2x + 1) dx = 12$ where $a > 0$, find a .

Answer: D

$a^2 + a = 12 \Rightarrow (a + 4)(a - 3) = 0 \Rightarrow a = 3$.

Question 6 INT-08 • Integration • spec MM7.4

Let $F(x) = \int_2^x (t^2 - 3t) dt$. Find $F'(3)$.

Answer: E

By the Fundamental Theorem of Calculus $F'(x) = x^2 - 3x$, so $F'(3) = 9 - 9 = 0$.

Question 7 INT-09 · Integration · spec MM7.1

Which of the following statements is/are true?

I $\int_{-2}^2 x^3 dx = 0$

II The area between $y = x^3$ and the x -axis from $x = -2$ to $x = 2$ is 0.

III $\int_{-2}^2 x^2 dx = 2 \int_0^2 x^2 dx$

Answer: F

I: odd function over a symmetric interval, so 0. II is false: the area is $2 \int_0^2 x^3 dx = 8$; the integral is 0 because the parts cancel. III: even function, true.

Question 8 INT-10 · Integration · spec MM7.2

Find $\int \left(x + \frac{1}{x}\right)^2 dx$.

Answer: A

$\left(x + \frac{1}{x}\right)^2 = x^2 + 2 + x^{-2}$, which integrates to $\frac{x^3}{3} + 2x - x^{-1} + c$. (Option E is a common error: you cannot integrate a power of a bracket that way.)

Question 9 INT-12 · Integration · spec MM7.2

Find the value of $\int_1^8 x^{-\frac{2}{3}} dx$.

Answer: C

$$[3x^{1/3}]_1^8 = 3(2 - 1) = 3.$$

Question 10 INT-13 · Integration · spec MM7.3

$f(x) = ax + b$ satisfies $\int_0^1 f(x) dx = 2$ and $\int_0^2 f(x) dx = 6$. Find $f(1)$.

Answer: E

$\frac{a}{2} + b = 2$ and $2a + 2b = 6$, so $a + b = 3$; subtracting the first equation from this gives $\frac{a}{2} = 1$, so $a = 2$, $b = 1$ and $f(1) = 3$.

Question 11 INT-16 · Integration · spec MM7.4

Given that $\int_0^4 f(x) dx = 10$, find $\int_0^4 (f(x) + 3x) dx - \int_4^0 f(x) dx$.

Answer: D

$$10 + 24 - (-10) = 44.$$

Question 12 INT-17 · Integration · spec MM7.6

The gradient of a curve at (x, y) is $3\sqrt{x}$ and the curve passes through $(4, 20)$. Find the y -intercept of the curve.

Answer: C

$$y = 2x^{3/2} + c; 20 = 16 + c \Rightarrow c = 4. \text{ The } y\text{-intercept is } 4.$$